



MONET Calc

Module of Numerical Exposure Time Calculator — NARIT LRS @ 2.4-m Thai National Telescope
User manual v6.5 · <https://monet-calc.streamlit.app> · [krittapas \[at\] narit.or.th](mailto:krittapas[at]narit.or.th)

Quick start (4 steps)

1. In the **sidebar**, define your target: pick a spectral template (or upload a 2-column ASCII table of wavelength [$\text{\AA}$] and f_{λ} , or build a continuum $\pm$ line), set redshift and Galactic E(B–V), then set the observed brightness (SDSS filter magnitude, monochromatic AB, or total flux). **2.** Set conditions and instrument: seeing, moon, clouds, target altitude, slit, CCD temperature, readout. **3.** Pick one of the three tabs and press its button. **4.** Read the plots and the color-coded warnings.

Calibration toggle (important): ON (default) scales predictions to match the measured 2026-04-01 commissioning performance of the as-built system ($\times 0.29$ with a measured color-dependent shape; conservative, realistic *now*). OFF gives the theoretical component model (use for forecasts after optics refurbishment).

The three calculators

Tab	Give	Get
① S/N $\leftrightarrow$ exposure time	N frames $\times$ exposure per frame — or — a target S/N at a chosen wavelength	S/N vs wavelength (per 4- $\text{\AA}$ pixel and per $\text{\AA}$), the expected spectrum in flux / ADU / S/N with warning shading — or — the total on-source time and per-frame split
② Simulated 1-D & 2-D spectra	N frames $\times$ exposure per frame	Expected 1-D spectrum (flux, ADU incl. peak pixel vs full well, S/N); raw 2-D frame in ADU (trace, airglow lines, slit vignetting, noise); reduced, stacked, sky-subtracted 2-D S/N map ($\lambda \times \text{arcsec}$) with realistic OH-band residuals; extended sources appear as a band of the extent you set, with sky fitted outside it
③ Recommended request time	N $\times$ t and whether nighttime calibration is needed	The time to REQUEST in the proposal: on-source + readout, slew, acquisition, thru-slit, operator, standard-star rounds (every 2 h), all $\div 0.65$ for the 35% weather margin; night calibration (1 h per half night) added outside the margin; daytime calibration shown as info, never charged

Warnings (shaded bands on the plots)

Color	Meaning	What to do
Red — saturation	peak pixel $\geq 65,535$ ADU in one frame	shorten frames, take more of them
Orange — non-linear	peak pixel above 95% of full well	shorten frames slightly
Purple — read-noise-dominated	RN^2 is the largest noise term	longer frames and/or slow readout
Brown — dark-dominated	dark current is the largest noise term	use the $-100\text{ }^{\circ}\text{C}$ operating point

Template guide — which template for which object

Templates are research-grade spectra from the standard ETC libraries (also used by ESO/Gemini/HST calculators). Names are **family_item**. All are renormalized to your requested brightness; only the *shape* matters. If a template does not cover part of 4000–8092 Å the app reports it and S/N there is undefined rather than extrapolated.

Stars — pickles_* (Pickles 1998 library)

Template(s)	Object type
pickles_o5v, b0v, b57v	hot O/B main-sequence stars (blue, Balmer + He lines); flux standards, young clusters
pickles_a0v, a5v	A dwarfs (strong Balmer absorption); Vega-like standards
pickles_f0v, f5v, g0v, g2v, g5v	F/G dwarfs; g2v = solar analog
pickles_k0v, k2v, k5v, m0v, m2v, m5v	K/M dwarfs (red, molecular bands); cool stars, exoplanet hosts
pickles_g8iii, k3iii, m0iii, m5iii	G/K/M giants; evolved stars, stellar populations of early-type galaxies
pickles_a0i, g2i, m2i	supergiants; luminous evolved massive stars

Galaxies — kc96_* (Kinney–Calzetti atlas)

Template(s)	Object type
kc96_elliptical	quenched / passive elliptical: red continuum, 4000-Å break, absorption lines only
kc96_bulge, kc96_s0	bulge-dominated and lenticular galaxies (note: red coverage ends ~7550 Å; see brown_* substitutes)
kc96_sa, kc96_sb, kc96_sc	spirals from early (Sa) to late (Sc) type: growing blue continuum and emission lines
kc96_starb1 ... starb6	starburst galaxies ordered by internal dust: starb1 = blue, nearly unobscured (young irregular-like); starb5/starb6 = dusty starbursts, E(B–V) up to ~0.7

Active nuclei — agn_*

Template(s)	Object type
agn_qso_ext (recommended)	luminous unobscured quasar; Francis+91 composite spliced with Türlér+99 — covers the full band at any redshift 0–4
agn_qso	same quasar composite, native version: rest coverage ends at 6000 Å — use only when redshift brings the rest-UV into the band ($z \approx 0.33$ –4)
agn_seyfert1	broad-line (type 1) Seyfert nucleus + host
agn_seyfert2, agn_ngc1068	obscured (type 2) AGN, narrow lines only; ngc1068 is the archetype with full-band coverage
agn_liner	low-ionization nuclear emission (LINER)

SDSS composites — dobos_* (Dobos+2012) and real-galaxy SEDs — brown_* (Brown+2014)

Template(s)	Object type
dobos_SF1 ... SF4	average SDSS star-forming galaxies, increasing star-formation activity; best for typical low- z emission-line galaxies (full band only to $z \approx 0.13$)
dobos_RED0, RED2, RED4	average SDSS passive (red-sequence) galaxies
dobos_BG, dobos_RG	blue-cloud and red-sequence population means
brown_NGC5033_sy1.5	real Seyfert 1.5 galaxy — full-band substitute for agn_seyfert1

Template(s)	Object type
brown_NGC4579_liner	real LINER — full-band substitute for agn_liner
brown_NGC3379_elliptical	real elliptical — full-band substitute for kc96_elliptical/bulge
brown_NGC4450_sab	real early-type (Sab) spiral
brown_NGC0628_sc	real late-type (Sc) spiral — full-band substitute for kc96_sc

Practical notes

Order of operations. The app applies your settings as: redshift → Galactic extinction → normalization. The magnitude you enter is therefore the *observed* brightness — what a catalog would list.

Redshift. Accepted to $z = 9$. Where the redshifted template has no rest-frame data (e.g., no X-ray/extreme-UV), the spectrum is padded with zeros and the app tells you which rest-frame regime is missing; S/N there is zero, and normalization in a padded region is refused. Kinney–Calzetti templates cover the full band to $z \approx 2.2$; the extended QSO to $z \approx 4$; SDSS composites only to $z \approx 0.13$.

Extended sources. Check “Extended source” to interpret brightness per arcsec² (uniform surface brightness; no slit losses; flux scales with slit × extraction area). Also set “Object extent along the slit” (new in v6.5): the extraction window matches the extent, the 2-D maps show the object as a seeing-blurred band of that size, and the sky is fitted only outside it — so the reduced map keeps your object instead of subtracting it as sky. If the object fills the whole slit, the app warns you and shows the map without sky subtraction: plan offset-sky exposures for such targets. For a compact but resolved object, instead keep a total magnitude and set “Source FWHM” — the profile widens to $\sqrt{(\text{seeing}^2 + \text{source}^2)}$, increasing slit losses realistically.

Slit choice. 1.8” gives $R \approx 750$, 2.7” → 500, 4.5” → 300 at 6000 Å. Under poor image quality (seeing worse than 2.5”) the wider slits recover large slit losses at modest resolution cost. The “moffat” PSF option models realistic seeing wings (10–25% more slit loss than Gaussian).

What to expect (calibrated mode, 1.8” slit, 1.5” seeing, dark sky, 30 min). Point source $r' \approx 18$: S/N ≈ 30 per pixel; $r' \approx 20$: S/N ≈ 7 ; practical continuum limit $r' \approx 21.5$ –22 for long stacks. Divide exposure times by ~ 3 when forecasting with the theoretical model instead.

Requesting time. Always quote tab ③’s “REQUEST” number in proposals — it includes all overheads, standard-star visits, and the 35% weather/mechanical margin. Daytime calibrations are free; nighttime calibration costs a fixed 1 h per half night and must be stated.

MONET Calc v6.5 · NARIT · source: github.com/krc2020/LRS_ETC · The empirical calibration is anchored to the 2026-04-01 BD+75°325 commissioning standard (shutter-corrected). Template provenance: Pickles 1998; Kinney et al. 1996; Calzetti et al. 1994; Francis et al. 1991; Türlér et al. 1999; Dobos et al. 2012; Brown et al. 2014.